

Lagrange Multiplier:

Property: The gradient of a function is perpendicular to the contour lines, the tangents to the contour lines of f & g are parallel iff the gradient of f & g are parallel.

Then we want points (x, y) where $g(x, y) = c$ & $\nabla f - \sum d_i \nabla g_i = 0$ where d is unrestricted sign.

Method of Lagrange multipliers (Equality constraints).

Given $u = f(x, y, z)$ subject to $g(x, y, z) = 0$.

Step 1: Write the auxiliary equation

$$F(x, y, z) = f(x, y, z) - d g(x, y, z).$$

Step 2: Find F_x, F_y & F_z by differentiating F partial w.r.t x, y , & z respectively.

Step 3: Solve the eqⁿs $F_x = 0, F_y = 0$ & $F_z = 0$ together with $\phi(x, y, z) = 0$.

Step 4: The values of x, y & z so obtained will give the stationary value of $f(x, y, z)$.

Note: Multiple constraints problem.

$u = f(x, y, z)$ subject to $\phi(x, y, z) = 0$ & $\psi(x, y, z) = 0$ then

Auxiliary eqⁿ $F(x, y, z) = f(x, y, z) - d_1 \phi(x, y, z) - d_2 \psi(x, y, z)$.

1. Find the maximum & minimum value of the function

$f(x, y) = 5x - 3y$ subject to the constraint $x^2 + y^2 = 136$

Solⁿ: Given $f(x, y) = 5x - 3y$, constraint $g(x, y) = x^2 + y^2 - 136 = 0$.

The auxiliary eqⁿ $F(x, y) = f(x, y) - d g(x, y)$

$$\Rightarrow F(x, y) = 5x - 3y - d(x^2 + y^2 - 136)$$

Diff this function w.r.t x & y

$$F_x = 5 - 2dx = 0 \Rightarrow x = 5/2d$$

$$F_y = -3 - 2dy = 0 \Rightarrow y = -3/2d$$

Substitute x & y into the constraints

$$(5/2d)^2 + (-3/2d)^2 = 136$$

$$\frac{25}{4d^2} + \frac{9}{4d^2} = 136$$

$$\Rightarrow \frac{34}{4d^2} = 136$$

$$\Rightarrow 34 = 544d^2$$

$$\Rightarrow d^2 = 1/16$$

$$\Rightarrow d = \pm \sqrt{1/16} = \pm 1/4$$

To find the points (x, y).

* Case (i) : $d = 1/4$

$$x = \frac{5}{2(1/4)} = 10, \quad y = \frac{-3}{2(1/4)} = -6 \quad \therefore \text{point } (10, -6)$$

* Case (ii) : $d = -1/4$

$$x = -10 \text{ \& } y = 6 \quad \therefore (-10, 6)$$

Substitute these values in given function

$$f(10, -6) = 5 \times 10 - 3 \times -6 = 68 \quad \text{maximum}$$

$$f(-10, 6) = 5 \times -10 - 3 \times 6 = -68 \quad \text{minimum}$$

4. Find the maximum & minimum values of $f(x, y, z) = 4y - 2z$ subject to the constraints $2x - y - z = 2$ & $x^2 + y^2 = 1$

Soln: $f(x, y, z) = 4y - 2z$, Let $g(x, y, z) = 2x - y - z - 2$, $h = x^2 + y^2 - 1 = 0$

The auxiliary equation

$$F(x, y, z) = f(x, y, z) - d_1 g(x, y, z) - d_2 h(x, y, z) \\ = 4y - 2z - d_1(2x - y - z - 2) - d_2(x^2 + y^2 - 1)$$

diff wrt x, y, z respectively & equate to zero

$$F_x = -2d_1 - 2xd_2 = 0 \text{ --- (1)}, \quad F_y = 4 + d_1 - 2d_2y = 0 \text{ --- (2)}, \quad F_z = -2 + d_1 = 0 \text{ --- (3)}$$

From (3) $d_1 = 2$ substitute d_1 in eqn (1) & (2)

$$-4 - 2d_2x = 0 \quad \& \quad 4 - 2d_2y = 0$$

$$\Rightarrow d_2x = -4/2 \quad \& \quad d_2y = 2$$

$$\Rightarrow d_2x = -2$$

$$\Rightarrow x = -2/d_2$$

$$\Rightarrow y = 2/d_2$$

Substitute x & y values in $h(x, y, z)$

$$4/d_2^2 + 4/d_2^2 = 1$$

$$\Rightarrow 8/d_2^2 = 1 \Rightarrow d_2^2 = 8 \Rightarrow d_2 = \pm \sqrt{8}$$

(2)

Case (8): $d_2 = \sqrt{13}$

$$x = -2/\sqrt{13} \quad \& \quad y = 3/\sqrt{13}$$

Substitute these values in $g(x, y, z)$

$$2(-2/\sqrt{13}) - 3/\sqrt{13} - z = 2$$

$$\Rightarrow z = \frac{-7}{\sqrt{13}} - 2$$

 $\therefore$ The stationary pt is $(-2/\sqrt{13}, 3/\sqrt{13}, \frac{-7}{\sqrt{13}} - 2)$

$$\therefore f(x, y, z) = 4(3/\sqrt{13}) - 2(\frac{-7}{\sqrt{13}} - 2) = 4 + 2\sqrt{13}$$

Case (9): $d_2 = -\sqrt{13}$

$$x = 2/\sqrt{13} \quad \& \quad y = -3/\sqrt{13}$$

$$\text{then } z = 2(2/\sqrt{13}) + 3/\sqrt{13} - 2 = 7/\sqrt{13} - 2$$

The stationary pt is $(2/\sqrt{13}, -3/\sqrt{13}, 7/\sqrt{13} - 2)$

$$\therefore f(x, y, z) = 4(-3/\sqrt{13}) - 2(7/\sqrt{13} - 2) = 4 - 2\sqrt{13}$$

Hence max: $4 + 2\sqrt{13}$ at $(-2/\sqrt{13}, 3/\sqrt{13}, \frac{-7}{\sqrt{13}} - 2)$ min: $4 - 2\sqrt{13}$ at $(2/\sqrt{13}, -3/\sqrt{13}, 7/\sqrt{13} - 2)$ 2. Find the maximum & minimum distances of the points $(3, 4, 12)$ from the sphere $x^2 + y^2 + z^2 = 1$

$$\text{soln: } f(x, y, z) = (x-3)^2 + (y-4)^2 + (z-12)^2$$

Subject to the constraint $g(x, y, z) = x^2 + y^2 + z^2 - 1 = 0$

The auxiliary eqn is

$$F(x, y, z) = (x-3)^2 + (y-4)^2 + (z-12)^2 - \lambda(x^2 + y^2 + z^2 - 1)$$

$$F_x = 2(x-3) - 2\lambda x = 0 \Rightarrow x = \frac{3}{1-\lambda}$$

$$F_y = 2(y-4) - 2\lambda y = 0 \Rightarrow y = \frac{4}{1-\lambda}$$

$$F_z = 2(z-12) - 2\lambda z = 0 \Rightarrow z = \frac{12}{1-\lambda}$$

Substitute these in constraint

$$\left(\frac{3}{1-\lambda}\right)^2 + \left(\frac{4}{1-\lambda}\right)^2 + \left(\frac{12}{1-\lambda}\right)^2 = 1$$

$$\frac{169}{(1-\lambda)^2} = 1 \Rightarrow (1-\lambda)^2 = 169 \Rightarrow 1-\lambda = \pm 13$$

Case (9): $1-\lambda = 13$, the point on the sphere is $(3/13, 4/13, 12/13)$

Case (ii) : $d = -13 \therefore (-3/13, -4/13, -12/13)$

$f(-3/13, -4/13, -12/13) = 14$

max. value is 14 at $(-3/13, -4/13, -12/13)$

min " " 12 at $(3/13, 4/13, 12/13)$.

8. $g(x, y, z) = 4x^2 + y^2 + 4z^2 - 16 = 0$ constraint

$f(x, y, z) = 8x^2 + 4yz - 16z + 600$

Auxiliary eqⁿ $F(x, y, z) = 8x^2 + 4yz - 16z + 600 - \lambda(4x^2 + y^2 + 4z^2 - 16)$

$F_x = 16x - 8\lambda x = 0 \Rightarrow 8x(2 - \lambda) = 0 \Rightarrow x = 0$ or $\lambda = 2$

$F_y = 4z - 2y\lambda = 0 \Rightarrow 2z = y\lambda$ — (2)

$F_z = 4y - 16 - 8z\lambda = 0 \Rightarrow y - 4 = 2\lambda z$ — (3)

Case (i) : $\lambda = 2$

From (2) $\Rightarrow z = y$, From (3) $y - 4 = 4y \Rightarrow y = \frac{-4}{3} = z$

Substitute these in constraint

$4x^2 + (-4/3)^2 + 4(-4/3)^2 = 16 \Rightarrow x^2 = 16/9 \Rightarrow x = \pm 4/3$

$\therefore$ The stationary pts are

$(-4/3, -4/3, -4/3)$ & $(4/3, -4/3, -4/3)$

$f = 8(16/9) + 4(16/9) - 16(-4/3) + 600 = 1928/3$ — (i)

Case (ii) : $x = 0$

Constraint becomes $y^2 + 4z^2 = 16$ — (4)

From (2) $\lambda = \frac{2z}{y}$

Substitute this in (3) ; $y - 4 = \frac{4z^2}{y} \Rightarrow y^2 - 4y = 4z^2$

From (4) $y^2 + (y^2 - 4y) = 16 \Rightarrow 2y^2 - 4y - 16 = 0$

$\Rightarrow (y - 4)(y + 2) = 0$

$y = 4$ & $y = -2$

(a) $y = 4, z = 0$: from (4)

The stationary pt is $(0, 4, 0)$; $f = 600$ — (ii)

(b) $y = -2, z = \pm\sqrt{3}$

$\therefore (0, -2, \sqrt{3})$ & $(0, -2, -\sqrt{3})$ are stationary pts.

$f(0, -2, \sqrt{3}) = 600 - 24\sqrt{3}$, $f(0, -2, -\sqrt{3}) = 600 + 24\sqrt{3}$ — (iii)

Compare all equations (i) (ii) & (iii)

max value is $1928/3$ at $(\pm 4/3, -4/3, -4/3)$

Yes, it is possible to find the point.

The maximum temperature occurs at the pts $(\pm 4/3, -4/3, -4/3)$

Karush-Kuhn-Tucker (KKT) optimality conditions:
(Inequality constraints).

For the problem:

Min $f(x)$ subject to $g_i(x) \leq b_i$
(n variables, m constraints)

The necessary conditions are:

$\nabla f - \sum d_i \nabla g_i = 0$ (Optimality)

feasibility: $g_i(x) - b_i \leq 0$

for $i = 1, 2, \dots, m$

$d_i (g_i(x) - b_i) = 0$ for $i = 1, 2, \dots, m$
(Complementary slackness condition)

$d_i \leq 0$ for $i = 1, 2, \dots, m$ (non-positivity)

Note: The non-positivity or non-negativity of d_i depends on the way the optimality condition is written.

Maximize $f(x)$

s.t. $g_i(x) \leq b_i$

the necessary conditions

$\nabla f / \nabla x - \sum_i d_i \nabla g_i / \nabla x = 0$

$d_i (g_i(x) - b_i) = 0$

$g_i(x) \leq b_i$

$d_i \leq 0$

Case: I

Max $f(x)$ s.t. $g_i(x) \leq b_i$

Case: II

Max $f(x)$ s.t. $g_i(x) \geq b_i$

Case: III

Min $f(x)$ s.t. $g_i(x) \leq b_i$

Case: IV

Min $f(x)$ s.t. $g_i(x) \geq b_i$

Max $f(x)$ subject to $g_i(x) \leq b_i$

The necessary conditions are

$\nabla f - \sum d_i \nabla g_i = 0$

$g_i(x) - b_i \leq 0$ for $i = 1, \dots, m$

$d_i (g_i(x) - b_i) = 0$ for $i = 1, \dots, m$

$d_i \geq 0$ for $i = 1, 2, \dots$ (non-negativity)

Maximize $f(x)$

s.t. $g_i(x) \leq b_i$

$\nabla f / \nabla x - \sum_i d_i \nabla g_i / \nabla x = 0$

$d_i (g_i(x) - b_i) = 0$

$g_i(x) \leq b_i$

$d_i \geq 0$

Case I: f is concave & g is convex

Case II: f is convex & g is concave

Case III: f is convex & g is convex

Case IV: f is concave & g is concave

① Maximize $Z = 12x_1 + 21x_2 + 2x_1x_2 - 2x_1^2 - 2x_2^2$ s.t. $x_1 + x_2 \leq 10$, $x_2 \leq 8$ & $x_1, x_2 \geq 0$ by deriving the KKT condition.

Soln: $Z = 12x_1 + 21x_2 + 2x_1x_2 - 2x_1^2 - 2x_2^2$
 Let $g_1 = x_1 + x_2 - 10 \leq 0$ & $g_2 = x_2 - 8 \leq 0$.

i) Optimality condition:

$$F(x_1, x_2) = 12x_1 + 21x_2 + 2x_1x_2 - 2x_1^2 - 2x_2^2 - d_1(x_1 + x_2 - 10) - d_2(x_2 - 8)$$

$$F_{x_1} = 12 + 2x_2 - 4x_1 - d_1 = 0 \quad \text{--- (a)}$$

$$F_{x_2} = 21 + 2x_1 - 4x_2 - d_1 - d_2 = 0 \quad \text{--- (b)}$$

ii) Complementary feasibility condition: $g_i(x) - b_i \leq 0$

$$g_1 = x_1 + x_2 - 10 \leq 0 \quad \text{--- (c)}, \quad g_2 = x_2 - 8 \leq 0 \quad \text{--- (d)}$$

iii) Complementary slackness condition: $d_i(g_i(x) - b_i) = 0$

$$d_1(x_1 + x_2 - 10) = 0 \quad \text{--- (e)} \quad \& \quad d_2(x_2 - 8) = 0 \quad \text{--- (f)}$$

iv) Non-negativity: $d_i \geq 0$.

Case (i): $d_1 = d_2 = 0$.

From (a) & (b) $12 + 2x_2 - 4x_1 = 0$ and $21 + 2x_1 - 4x_2 = 0$

Solve these eqns. we get $x_1 = 7.5$, $x_2 = 9$

But $x_2 = 9 \neq 8$. It is infeasible.

Case (ii): $d_1 = 0$, $d_2 \neq 0$

From (f), $x_2 = 8$.

Substitute x_2 value in eqn (a), $12 + 16 - 4x_1 = 0 \Rightarrow x_1 = 7$.

But $x_1 + x_2 = 7 + 8 = 15 \neq 10$, this violates $g_1 \leq 10$.

$\therefore$ Infeasible.

Case (iii): $d_1 \neq 0$, $d_2 = 0$

From (e), $x_1 + x_2 - 10 = 0 \Rightarrow x_2 = 10 - x_1$

From optimality condition,

$$12 + 2(10 - x_1) - 4x_1 - d_1 = 0 \Rightarrow 32 - 6x_1 = d_1$$

$$21 + 2x_1 - 4(10 - x_1) - d_1 = 0 \Rightarrow 6x_1 - 19 = d_1$$

$$\therefore x_1 = 4.25 \quad \& \quad x_2 = 5.75 \quad \therefore d_1 = 6.5$$

$$x_1 + x_2 \leq 10, \quad x_2 \leq 8 \quad \text{--- this is feasible}$$

Case (iv): $d_1 \neq 0$, $d_2 \neq 0$.

From (c) & (f), $x_2 = 8$, $x_1 + x_2 - 10 = 0 \Rightarrow x_1 = 2$

again check the constraint. It satisfies all condition $\therefore$ feasible

∴ The maximum value is

$$Z = 118.375 \quad \text{at} \quad x_1 = 20.25, \quad x_2 = 50.75$$

$$Z = 184 \quad \text{at} \quad x_1 = 2, \quad x_2 = 8.$$

satisfied.

$f(0,0,0) = 0$ minimum at $x_1, x_2, x_3 \geq 0$.

3. Minimize $f = x_1^2 + x_2^2 + 60x_1$, subject to $x_1 \geq 80$, $x_1 + x_2 \geq 120$ by deriving the KKT conditions.

Solⁿ: $f = x_1^2 + x_2^2 + 60x_1$,

Let $g_1 = 80 - x_1 \leq 0$ & $g_2 = 120 - x_1 - x_2 \leq 0$.

A.E. is $F(x_1, x_2) = x_1^2 + x_2^2 + 60x_1 - d_1(80 - x_1) - d_2(120 - x_1 - x_2)$

To verify KKT conditions:

i) Optimality condition:

$$F_{x_1} = 2x_1 + 60 + d_1 + d_2 = 0 \Rightarrow x_1 = \frac{-d_1 - d_2 - 60}{2}$$

$$F_{x_2} = 2x_2 + d_2 = 0 \Rightarrow x_2 = -d_2/2$$

ii) Feasibility condition: $g_1 = 80 - x_1 \leq 0$, $g_2 = 120 - x_1 - x_2 \leq 0$.

iii) Complementary slackness condition: $d_i(g_i(x) - b_i) = 0$

$$d_1(80 - x_1) = 0, \quad d_2(120 - x_1 - x_2) = 0$$

iv) Non-positivity: $d_i \leq 0$. i.e., $d_1 \leq 0$, $d_2 \leq 0$.

Case Analysis:

Case (i): $d_1 = d_2 = 0$

From (i) $x_1 = -30$, $x_2 = 0$

Substitute x_1 & x_2 in (ii). $g_1 = 80 + 30 \neq 0$, $g_2 = 120 + 30 - 0 \neq 0$

$\therefore$ Infeasible.

Case (ii): $x_1 = 80$, $d_2 = 0$ ($d_2 = 0, d_1 \leq 0$)

From (i): $x_2 = 0$,

Check feasibility: $(80, 0)$ violates the second constraint $80 + 0 \geq 120$ (Infeasible)

Case (iii): $d_1 = 0$, $x_1 + x_2 = 120$ ($d_1 = 0, d_2 \leq 0$)

From (i) $d_2 = -2x_2$ & $2x_2 + (120 - x_2) = \frac{2x_2 - 60}{2}$

$$\Rightarrow 120 - x_2 = x_2 - 30 \Rightarrow x_2 = 75, \therefore x_1 = 120 - 75 = 45$$

Check feasibility: $x_1 = 45 \not\geq 80$ (Infeasible).

$$\therefore f(80, 40)$$

Case (iv): $x_1 = 80$, $x_1 + x_2 = 120$ ($d_1 \neq d_2 \neq 0$)

$$\text{fmin} = 12800.$$

$$\therefore x_2 = 120 - 80 = 40.$$

Substitute these in (iii) $d_2 = -80$, $d_1 = -140$. $\therefore d_i \leq 0$.

Unconstrained Optimization

1. Newton's Method to optimize quadratic functions.

Consider the quadratic approximation of the function $f(x)$ at $x = x_i$ using the Taylor's series expansion

$$f(x) \approx f(x_i) + (x - x_i)^T [\nabla f_i] + \frac{(x - x_i)^T [H_i] (x - x_i)}{2} \quad \text{--- (1)}$$

where $[\nabla f_i]$ is $[\partial f / \partial x_1, \partial f / \partial x_2, \partial f / \partial x_3, \dots]$ & $[H_i]$ is the Hessian matrix of f evaluated at the point x_i .

By setting the partial derivatives to zero for the minimum of $f(x)$, we obtain $\frac{\partial f(x)}{\partial x_j} = 0, j = 1, 2, \dots, n$ --- (2)

$$\text{From (1) \& (2)} \quad [\nabla f]^T = [\nabla f_i]^T + [H_i] (x - x_i)^T = 0 \quad \text{--- (3)}$$

If $[H_i]$ is non-singular, eqⁿ (3) can be solved to obtain an improved approximation ($x = x_{i+1}$) as

$$x_{i+1} = x_i - [H_i]^{-1} \cdot [\nabla f_i]^T \quad \text{--- (4)}$$

Since higher order terms have been neglected in eqⁿ (1), eqⁿ (4) is to be used iteratively to find the optimum solution x^* .

Algorithm:

Step 1: Find ∇f and substitute initial ~~condition~~ value (i.e., ∇f_i)

Step 2: Compute Hessian matrix & H^{-1}

Step 3: Use Newton's method to find new point

$$x_{i+1} = x_i - H^{-1} \nabla f_i$$

Step 4: Check the optimum of x_{i+1} by $\nabla f(x_{i+1}) \cong 0$.

1. minimize $f(x_1, x_2) = x_1^2 + 4x_2^2$ by taking the starting point (6)

as $x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Soln: Given $f(x_1, x_2) = x_1^2 + 4x_2^2$

$$* \nabla f = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{bmatrix} = \begin{bmatrix} 2x_1 \\ 8x_2 \end{bmatrix}$$

at the starting point $x_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$, $\therefore \nabla f_{\text{start}} = \begin{bmatrix} 2 \\ 8 \end{bmatrix}$

$$* H = \begin{bmatrix} \partial^2 f / \partial x_1^2 & \partial^2 f / \partial x_1 \partial x_2 \\ \partial^2 f / \partial x_2 \partial x_1 & \partial^2 f / \partial x_2^2 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 8 \end{bmatrix} \Rightarrow H^{-1} = \begin{bmatrix} 0.5 & 0 \\ 0 & 0.125 \end{bmatrix}$$

By Newton's method $x_{i+1} = x_i - H^{-1} \nabla f_i$

$$x_2 = x_1 - H^{-1} \nabla f_1$$

$$x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.5 & 0 \\ 0 & 0.125 \end{bmatrix} \begin{bmatrix} 2 \\ 8 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Verify the minimum pt at $x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$

$$\therefore \nabla f(0,0) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\therefore (0,0)$ is the stationary point

Since the hessian matrix is ^{strictly} positive definite because all eigen values are ^{strictly} positive. Hence the critical point is local minimum

Therefore $f(0,0) = 0$

2. Minimize $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$ by taking the starting point as $x_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

Soln: $\nabla f = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} \Rightarrow \nabla f_{(0,1)} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

$$H = \begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix} \Rightarrow H^{-1} = \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$

To find next approximation by Newton's method

$$x_2 = x_1 - H^{-1} \nabla f_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix} - \begin{bmatrix} 0.5 & -0.5 \\ -0.5 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1.5 \end{bmatrix}$$

Verify the optimum point at $x_2 = \begin{bmatrix} -1 \\ 1.5 \end{bmatrix}$

$$\nabla f = \begin{bmatrix} 1 - 4 + 2 \times 1.5 \\ -1 - 2 + 2 \times 1.5 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \nabla f \Rightarrow 0$$

$$\therefore x_2 = \begin{bmatrix} -1 \\ 1.5 \end{bmatrix} \text{ is stationary point}$$

Nature of stationary point

$$H = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix} \Rightarrow d^2 - 6d + 4 = 0 \quad \text{H is +ve definite}$$

$\therefore$ The stationary point is minimum point

$$f(-1, 1.5) = -5/4 = -1.25$$

4. Minimize $f(x) = x_1^2 - x_1 x_2 + 3x_2^2$ starting at $x^0 = [1, 2]^T$

$$\text{Soln: } \nabla f = \begin{bmatrix} 2x_1 - x_2 \\ -x_1 + 6x_2 \end{bmatrix} \Rightarrow \nabla f, \text{ at the point}$$

$$\nabla f_1 = \begin{bmatrix} 0 \\ 11 \end{bmatrix}$$

$$H = \begin{bmatrix} 2 & -1 \\ -1 & 6 \end{bmatrix} \Rightarrow H^{-1} = \begin{bmatrix} 0.5454 & 0.0909 \\ 0.0909 & 0.1818 \end{bmatrix}$$

$$x_2 = x_1 - H^{-1} \nabla f_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 0.5454 & 0.0909 \\ 0.0909 & 0.1818 \end{bmatrix} \begin{bmatrix} 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$\therefore (0, 0)$ is the stationary point

H is +ve definite. Hence the stationary point is minimum

$$f(0, 0) = 0.$$

5. Minimize $f(x, y) = [1 + (x_1 + x_2 - 5)^2] \cdot [1 + (3x_1 - 2x_2)^2]$ taking initial points $(10, 10)$ & $(2, 2)$. (Not in your syllabus).

$$\nabla f =$$

Fibonacci Method :

The Fibonacci search is based on the sequence of Fibonacci numbers which are defined by the equations

$$F_0 = 1, F_1 = 1, F_{N+1} = F_N + F_{N-1} \text{ for } N = 1, 2, \dots$$

Then the Fibonacci numbers are 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

Algorithm :

1. Determine the no of function evaluations, $N-1$
2. Divide the interval $[a_1, b_1]$ evenly into F_N subintervals and hence the length of each subinterval is $\frac{1}{F_N}(b_1 - a_1)$
3. In the first iteration, the grid points are

$$x'_1 = b_1 - \frac{F_{N-k}}{F_{N-k+1}}(b_1 - a_1) \quad \& \quad x'_2 = a_1 + \frac{F_{N-k}}{F_{N-k+1}}(b_1 - a_1) \quad k=0, 1, 2, \dots$$

Compare the function values at these two points $(f(x'_1), f(x'_2))$ & decide the new interval.

→ If $f(x'_1) > f(x'_2)$ then the new interval is $[x'_1, b]$

→ If $f(x'_1) < f(x'_2)$ then " " " " $[a, x'_2]$

4. Repeat the steps 3 for other k values to obtain $x'_1, x'_2, \dots$

6. After completing all prescribed iterations, determine the approximate value of x as the midpoint of the final interval obtained.

Note : * Grid point $P_i = a_1 + \frac{i}{F_N}(b_1 - a_1)$.

* If a minimizer has to be found with tolerance (level of uncertainty) ϵ , then the length of the last interval

$$\frac{b-a}{F_N} < \epsilon.$$

1. Apply Fibonacci search method to find the value of x that minimizes $f(x) = x^4 - 14x^3 + 60x^2 - 70x$ over the range $[0, 2]$ by performing 7 iterations

Solⁿ: For $N = 7$ iterations we need the Fibonacci numbers upto F_7 .

WKT $F_0 = 1, F_1 = 1, F_2 = 2, F_3 = 3, F_4 = 5, F_5 = 8, F_6 = 13, F_7 = 21$

Given $[a, b] = [0, 2]$ $\therefore$ Length of the interval $b - a = 2$.

First Interior points: ($k=1$)

$$x_1' = b - \frac{F_{N-k}}{F_{N-k+1}} (b-a)$$

$$x_2' = a + \frac{F_{N-k}}{F_{N-k+1}} (b-a)$$

$x_1' = 2 - \frac{F_6}{F_7} (2) = 2 - \frac{13}{21} (2)$, $x_2' = 0 + \frac{F_6}{F_7} (2) = 0 + \frac{13}{21} (2)$

$x_1' = 0.7619$, $x_2' = 1.2381$

$\therefore f(0.7619) = -24.3584$, $f(1.2381) = -18.9140$

Since $f(x_1') < f(x_2')$, the new interval is $[0, x_2']$ i.e., $[0, 1.2381]$

Second Iteration: $k=2$, $b_1 = 1.2381$, $a_1 = 0$.

$$x_1^2 = 1.2381 - \frac{F_5}{F_6} (1.2381)$$

$$x_2^2 = 0 + \frac{F_5}{F_6} (1.2381)$$

$$= 1.2381 - \frac{8}{13} (1.2381)$$

$$= \frac{5}{13} (1.2381)$$

$$= 0.4762$$

$$= 0.7619$$

$\therefore f(0.4762) = -21.1884$, $f(0.7619) = -24.3584$

Since $f(x_1^2) > f(x_2^2)$, the new interval is $[0.4762, 1.2381]$

Third Iteration: $k=3$, $b_1 = 1.2381$, $a_1 = 0.4762$

$$x_1^3 = 1.2381 - \frac{F_4}{F_5} (1.2381 - 0.4762)$$

$$x_2^3 = 0.4762 + \frac{F_4}{F_5} (1.2381 - 0.4762)$$

$$= 1.2381 - \frac{5}{8} (0.7619)$$

$$= 0.4762 + \frac{5}{8} (0.7619)$$

$$= 0.7619$$

$$= 0.9524$$

$f(0.7619) = -24.3584$, $f(0.9524) = -23.5157$

$\therefore f(x_1^3) < f(x_2^3) \Rightarrow$ the new interval is $[0.4762, 0.9524]$

Fourth Iteration: $k=4$, $b_1 = 0.9524$, $a_1 = 0.4762$

$$x_1^4 = 0.9524 - \frac{F_3}{F_4} (0.9524 - 0.4762)$$

$$x_2^4 = 0.4762 + \frac{F_3}{F_4} (0.9524 - 0.4762)$$

$$= 0.6667$$

$$= 0.7619$$

$f(x_1^4) = -23.9509$, $f(x_2^4) = -24.3584$

$\therefore f(x_1^4) > f(x_2^4) \Rightarrow [0.6667, 0.9524]$

Fifth Iteration: $k=5$

$$x_1^5 = 0.7619, \quad x_2^5 = 0.8571$$

$$f(0.7619) = -24.3584, \quad f(0.8571) = -24.1951$$

$$f(x_1^5) < f(x_2^5) \therefore [0.6667, 0.8571]$$

Sixth Iteration: $k=6$

$$x_1^6 = 0.7619, \quad x_2^6 = 0.7619$$

$$\therefore f(x_1^6) = f(x_2^6) = f(0.7619) = -24.3584 \text{ with } [0.6667, 0.8571]$$

$$\therefore \text{The mid point } x_{\min} = \frac{0.6667 + 0.8571}{2} = 0.7619$$

2. Apply Fibonacci method to minimize $f(x) = x^2 - 1.5x$ in $[0, 1]$ within the interval of uncertainty 0.25 of the ^{initial} interval

Solⁿ: ~~known~~. The initial interval length $b-a=1$.

$$\text{required final interval length } \frac{1}{F_N}(b-a) < 0.25 \Rightarrow \frac{1}{F_N} < 0.25$$

$$\Rightarrow F_N > 4$$

The smallest Fibonacci no greater than 4 is $F_4 = 5$

hence $N=4$, Let $F_0=1, F_1=1, F_2=2, F_3=3, F_4=5$ ~~then~~

I Iteration ($k=1$):

$$x_1^1 = b - \frac{F_{N-k}}{F_{N-k+1}}(b-a) \text{ and } x_2^1 = a + \frac{F_{N-k}}{F_{N-k+1}}(b-a)$$

$$x_1^1 = 0.4$$

$$x_2^1 = 0.6$$

$$f(0.4) = -0.44$$

$$f(0.6) = -0.54$$

$$f(0.4) > f(0.6) \therefore [0.4, 1]$$

II Iteration ($k=2$):

$$x_1^2 = 0.6, \quad x_2^2 = 0.8$$

$$f(0.6) = -0.54 > f(0.8) = -0.56 \therefore [0.6, 1]$$

III Iteration ($k=3$):

$$x_1^3 = 0.8, \quad x_2^3 = 0.8, \quad f(0.8) = -0.56$$

$$\therefore f_{\min} = -0.56 \text{ at } [0.6, 1]$$

$$\text{The mid point } x^* = \frac{0.6+1}{2} = 0.8$$

$$f(x^*) = f(0.8) = -0.56$$